

TOP-LEVEL SAFETY CLAIM (C_{top})

"The Physical AI Agent operates acceptably safe within Operational Design Domain $\mathcal{X}_{ODD}$."

ARGUMENT A_1

Fault Containment

All single-point hardware and software crashes are deterministically arrested by the MCU safety enforcer within 50 ms.

EVIDENCE E_1

HIL Fault Log (REL-01)

- 1000/1000 faults contained
- 0 collisions recorded
- Max stopping time: 42 ms
- Watchdog trip: 50 ms

ARGUMENT A_2

Latency Freshness

The $P_{99.9}$ sense-to-actuation latency Δt_{wall} remains strictly bounded so stopping distance $d_{stop} \leq d_{gap}$.

EVIDENCE E_2

Metrology CDF (REQ-01)

- $P_{50} = 22$ ms
- $P_{99.9} = 72$ ms < 80 ms
- Margin: +18 cm clearance
- Zero seqlock overruns

ARGUMENT A_3

Human Authority

Human operators retain un-preemptible authority via bumpless joystick override and dedicated hardware E-stop circuits.

EVIDENCE E_3

Override Logs (AUTH-01)

- 50/50 seamless takeovers
- Peak jerk < 15 rad/s³
- Zero gearbox shock damage
- Tamper-evident hash log

ARGUMENT A_4

ODD Enforcement

Real-time interoceptive and exteroceptive health monitors detect out-of-ODD transitions and trigger safe deceleration.

EVIDENCE E_4

ODD Monitor Logs

- Dark/glare detection < 20 ms
- Friction drop fallback
- Thermal derating at 85°C
- Category 1 stop verified

THE 3 ACCOUNTABLE RELEASE VERDICTS

DEPLOY (All evidence thresholds met; unconstrained ODD) | **CONDITION** (Restricted speed / active human supervisor) | **REFUSE** (Gate failure; deployment blocked)